

Context-Aware Banking Customer Support AI Agent

Project Overview

The Context-Aware Banking Customer Support AI Agent is a Python-based conversational system designed to simulate an intelligent banking support assistant.

The agent understands user intent, maintains simple conversation memory, generates support tickets, and retrieves ticket status using rule-based Natural Language Processing techniques.

Problem Statement

Customers often need quick support for common banking issues such as debit card problems, ticket updates, or general assistance.

This project builds an AI-style agent capable of:

- Understanding user messages
- Classifying intent
- Maintaining basic user memory
- Creating and tracking support tickets
- Providing contextual responses

Key Features

- Intent classification (positive, negative, query)
- Context-aware conversation memory
- Smart introduction using single-word name detection
- Automatic ticket generation
- Ticket status lookup using a database file
- Router-based modular agent architecture
- Clean command-line conversational interface

Agent Architecture

User Input → Router() → Smart Intro Agent → Context Agent → Intent Classifier → Response Handlers

◆ Core Components

- router() – Main decision engine that directs messages.
- smart_intro() – Detects user's name and stores it in memory.
- classify_message() – Identifies message intent.
- feedback_handler() – Generates responses for positive/negative messages.
- query_handler() – Retrieves ticket status from database.
- context_agent() – Handles context-aware questions like latest ticket.

Project Structure

Banking_AI_Agent_Project/

- agents.py Main AI agent logic
- database.py Ticket database
- README.md Project documentation

Technologies Used

- Python 3.x
- Random module (ticket generation)
- Regular Expressions (pattern matching)
- Rule-based NLP logic
- Modular Function Design

How to Run the Project

1. Open terminal in the project folder.
2. Run: `python agents.py`
3. Start chatting with the agent.

Type 'exit' to stop the conversation.

Example :

You: Sneha

Agent: Nice to meet you, Sneha! How can I assist you today?

You: I have an issue with my debit card

Agent: We apologize for the inconvenience. Ticket #398472 has been created.

You: what is my latest ticket?

Agent: Your most recent ticket number is #398472.

You: thank you

Agent: Thank you for your kind words, Sneha!

Future Enhancements

- Replace rule-based classifier with ML/NLP model
- Add sentiment analysis
- Connect to real banking APIs
- Store ticket history using a database (SQLite/MySQL)
- Build GUI or web interface

Educational Value

This project demonstrates AI Agent Design Concepts, Context Handling in Conversations, Intent Routing Architecture, and Modular Python Programming.